

Next Best Action / Next Best Offer Recommendation for Insurance

Executive Summary

An ML-powered personalization solution that identifies and ranks the most relevant insurance product, rider, renewal action, service interaction, or digital engagement action for an eligible policyholder or prospect. The solution uses customer profile, policy portfolio, premium behavior, claims patterns, digital engagement, and historical campaign outcomes to estimate propensity and produce controlled recommendations.

Business Problem

- Generic renewal and cross-sell campaigns often have low relevance and weak conversion.
- Policyholders may not receive the product, rider, or service action most relevant to their lifecycle stage or protection gap.
- Business teams need a measurable, data-driven personalization capability across agent, web, app, and contact-center channels.
- ML recommendations must remain subject to underwriting constraints, eligibility rules, compliance requirements, and product governance.

Objectives

- Predict the probability that a customer will respond to each candidate insurance offer or action.
- Rank eligible candidates and return Top-N recommendations.
- Improve conversion, renewal retention, and digital engagement while reducing campaign waste.
- Provide explainable recommendation drivers and measurable business KPIs.
- Expose recommendations through APIs for agent portals, mobile apps, web journeys, and service workflows.

Scope

- Customer and policy-profile ingestion
- Premium, claims, and engagement feature engineering
- Offer and action propensity modeling
- Candidate generation and eligibility filtering
- Recommendation ranking and API delivery
- Feedback capture, monitoring, and retraining

Key Use Cases

- Motor policy add-on recommendation
- Family floater health policy recommendation
- Home insurance recommendation
- Travel insurance recommendation
- Critical illness rider recommendation
- Renewal retention action
- Wellness app activation or self-service action

ML / Analytics Approach

- Start with Logistic Regression as an interpretable propensity baseline.
- Compare tree-based models once the baseline is measured.
- Model the customer + offer + observation-date unit so each candidate has a response probability.
- Use time-aware validation where production data supports it.
- Calibrate probabilities if they are used for ranking or thresholding.
- Apply underwriting, portfolio, and compliance filters before final recommendation.
- Capture actual outcomes for retraining and monitoring.

Success Metrics

- Precision@K and Recall@K
- ROC-AUC / PR-AUC
- Quote-to-bind or offer acceptance rate
- Renewal retention uplift
- Recommendation coverage
- Digital engagement rate
- Incremental uplift against a control group

Risks and Controls

- Historical campaign selection bias can distort propensity estimates.
- Popular products can dominate recommendations without coverage and fairness controls.
- Cold-start customers may require fallback business rules.
- Incorrect policy or claims data can degrade recommendations.
- Privacy, consent, and insurance regulation requirements must be built into implementation.

High-Level Architecture

Channels -> API Gateway -> Customer / Policy / Claims Data -> Feature Engineering -> Candidate Generation and Eligibility -> ML Propensity Model -> Ranking Engine -> Recommendation API -> Agent / Web / App -> Feedback and Monitoring

Expected Deliverables

- Business requirements and use-case catalogue
- Data dictionary and prototype or synthetic dataset
- EDA and feature-engineering pipeline
- Model training and evaluation pipeline
- Recommendation API
- Channel integration prototype
- Monitoring, governance, and technical documentation